

LOAD CALCULATOR PRO

Part of the Field Hospital Calculator suite. Estimates generator load (kW/kVA), peak starting kVA, and fuel runtime for field hospital and similar sites. Supports editable equipment lists, custom rows, and capacity checking against an actual generator size.

How to run

Open 'index.html' in a modern browser. No server required. Works offline.

Dependencies: 'equipment-data.js' (category and item definitions), 'script.js' (main logic), 'styles.css'. Equipment data is loaded from 'LOAD_CALC_PRO_EQUIPMENT' (in 'equipment-data.js' or inline).

Functionality

Equipment categories and table

- Categories: Standard Medical, Emergency/Critical, Office & IT, HVAC/Hygiene/Work Lights (data-driven from equipment-data).
- Per row: Item name, Qty, kW each, PF, kW total, kVA Peak. Editable kW and PF; quantity drives totals.
- Search ??? Filter equipment by name (live filter). Sidebar shows all-items totals plus filtered-view subtotals (kW and kVA) when search is active.
- Sort ??? Name A???Z/Z???A, kW High/Low, kVA Peak High/Low (persisted in localStorage).
- Custom rows ??? Add items with name, kW, PF, qty via the form above each category.

Delete via row icon.

- Reset Qty (per category) and Reset all quantities ??? Zero out quantities; Clear Sheet resets quantities, kW/PF to defaults, and generator/fuel inputs.

Summary and capacity

- Total running kW (all items) ??? Sum of (qty ?? kW) across all rows (master planning total).
- Total running kVA (all items) ??? Sum of (qty ?? kW ?? PF) across all rows.
- Filtered view load (kW/kVA) ??? Subtotals across visible rows only; shown when search is active for focused analysis.
- Peak starting kVA ??? Largest motor start (4?? running kVA for inductive loads, PF < 0.95) plus running kVA of the rest. Drives minimum generator rating.
- Recommended generator size (kVA) ??? $\max(\text{Peak kVA}, \text{Total running kVA} \times 0.8)$.
- Generator capacity check ??? Enter actual generator kVA; see kVA margin and status pill (Adequate margin / Tight / Insufficient / Peak start fail).

Fuel and runtime

- Fuel tank capacity (Gallons) and Consumption (Gal/hr per kW) ??? Used to estimate Approximate runtime (hours).

Scenarios

- Save scenario ??? Saves current equipment quantities, kW/PF, custom rows, and sidebar inputs (kVA, fuel, rate) to browser storage. Validation must pass for key inputs before saving.
- Load / Delete / Clear all ??? Restore or remove saved scenarios from dropdown.
- Import from file / Export to file ??? Export format: JSON (backup/re-import) or CSV (human-readable report). Import accepts JSON scenario files.
- Worksheet autosave/restore ??? In-progress worksheet state autosaves in this browser and restores on reopen.

- Clear Autosaved State (toolbar) ??? Removes worksheet autosave data only; named saved scenarios remain.
- Last autosaved (toolbar) ??? Shows the timestamp of the most recent worksheet autosave in this browser.

Validation and UX

- Input validation ??? Sidebar (available kVA, fuel capacity, fuel rate) and table inputs (qty, kW, PF) validated on blur with red border and inline message. Save blocked when errors exist.
- Placeholder behavior ??? Number inputs use placeholders (e.g. 0); focus clears ???0??? for easier entry. Resets and load set empty where value is 0 so placeholders show.

Other

- Print ??? Print Summary; toolbar, scenario panel, sort bar, search, and add-row forms are hidden in print.
- Reset Worksheet ??? Full reset: quantities, kW/PF to original defaults, remove custom rows, clear generator/fuel inputs and search.

Calculation method and assumptions

(For electrical engineer review: formulas and constants used for load and generator sizing.)

Running load

- Per row: Running kVA = qty ?? (kW ?? PF). Running kW = qty ?? kW.
- Totals: Total running kW and total running kVA are the sum over all equipment rows. Rows with qty = 0 contribute nothing.

Peak starting kVA

- Inductive loads: A row is treated as inductive (motor / non-linear) when $PF < 0.95$ and $kW > 0.1$. For these, starting kVA per unit is taken as running kVA per unit $\times 4$ (motor start factor).
- Largest motor for peak: Only rows with qty > 0 are considered. Among those, the row with the largest single-unit starting kVA (inductive: $4 \times$ running kVA per unit; otherwise running kVA per unit) is the ???largest motor.???
- Peak kVA formula:
Peak kVA = (running kVA of all other rows, with the largest-motor row contributing only (qty $\times 1$) $\times$ running kVA per unit) + (starting kVA of one unit of the largest motor).
If total running kW = 0, peak kVA = 0. Peak kVA is never less than total running kVA.
- Constants: Motor start factor = 4.0 (conservative typical inrush assumption; not from a specific standard). No diversity factor is applied to the peak.

Recommended generator size

- Recommended kVA = $\max(\text{Peak kVA}, \text{Total running kVA} \times 0.8)$. The divisor 0.8 is a continuous-load safety factor (equivalent to 80% continuous loading).

Fuel runtime

- Runtime (hours) = Fuel tank capacity (gallons) ?? (Total running kW ?? Consumption in gal/h per kW).
- If total kW, capacity, or consumption is 0 or missing, runtime is shown as 0.

Summary for reviewer

- kVA = kW ?? PF throughout; $PF < 0.95$ and $kW > 0.1$ used only to apply the $4 \times$ start factor.
- Peak uses only equipment with qty > 0; 0-qty rows do not affect peak or recommended kVA.
- This tool is for estimation and planning. Final design and generator selection should be verified by a qualified electrical engineer.

UX/UI design

- Layout: Banner ??? Toolbar (Clear Autosaved State, Last autosaved, Print, Reset Quantities, Reset Worksheet) ??? Scenarios ??? Sort bar ??? Search bar ??? Table note ??? Main grid (categories column + sidebar). Sidebar: load summary, generator capacity check, fuel & runtime.
- Suite alignment: Same banner, scenario block pattern, and button roles (Print blue, Save/Load green, Delete amber, Clear red, Secondary gray). Collapsible categories; status pills for capacity.
- Files: 'index.html', 'styles.css', 'script.js', 'equipment-data.js', 'UNIFIED_UX_SPEC.md' (suite style guide).

File structure

File	Purpose
'index.html'	Page structure, sidebar, scenario UI, sort/search.
'script.js'	Build tables from equipment data, recalc, validation, scenarios, search filter, sort.
'styles.css'	Layout, cards, table, validation and placeholder styles, print.
'equipment-data.js'	'LOAD_CALC_PRO_EQUIPMENT' ??? categories and items (name, kw, pf).
'UNIFIED_UX_SPEC.md'	Unified UX/UI spec for the calculator suite.

Technical notes

- Sizing: See Calculation method and assumptions above. In short: running kVA = qty ?? (kW ?? PF); peak = running kVA excluding largest motor + (one unit of largest motor at 4??); only rows with qty > 0 are considered for largest motor; recommended kVA = max(Peak, Total kVA ?? 0.8).
- Storage: Worksheet autosave under 'loadCalcProScenario'; autosave timestamp under 'load-pro-lastSaved'; named scenarios under 'loadCalcProScenarios'; sort preference under 'loadCalcProSort'.
- Search: Rows without a name match get class 'search-hidden'; recalc sums only visible rows.

Tooltip/help notes

- Scenarios help explains save/load/delete/clear/import/export and scenario naming guidance.
- Equipment list help (next to Sort) explains list focus tools and filtered-view subtotal behavior.
- Summary/capacity/fuel helps clarify all-items calculation basis and planning assumptions.

Use for estimation only. Verify with a qualified electrical engineer for final design.